RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

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III Semester B.E. Regular / Supplementary Examination February/March 2026

COMMON TO EC / EE / EI / ET

Analysis and Design of Digital Circuits with HDL

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

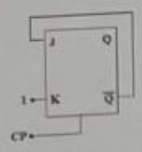
1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

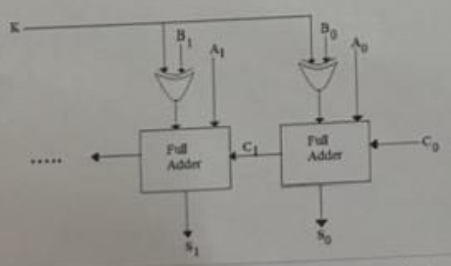
Question No	Question	M	CO	BT
1.1	Identify the value of Y in the following expression: a = 4'b1001; Y = &a .	01	1	1

1.2	For the following level sensitive cyclic behavior the synthesis tool will infer _____hardware.
	
	module test(input en, input [3:0]a, output [3:0] y);
	reg[3:0] y;
	always @ (en,a)
	if(en) y=a;
	else y=4'bZ;
	endmodule
		02	4	3

1.3	Initial block will execute _____ time during simulation.	01	1	1
1.4	An input to a module is 8 bit data , write a Verilog port declaration for this 8 bit input.	01	1	2
1.5	Write the number of Full adders required to design a 16-bit ripple carry adder.	01	1	2

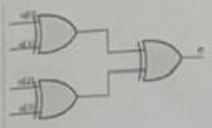
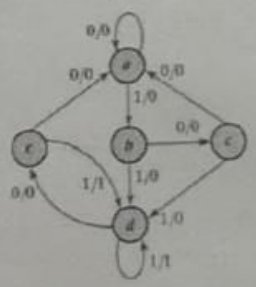
1.6	Write Verilog Code for a 32-bit wide tri-state buffer using data flow modelling.	02	3	3
1.7	In the JK flip-flop in figure , Assume we have J = Q' and K = 0, and Q is initially cleared to 0. After applying six clock pulses to this circuit, the sequence at the Q output will be 	02	4	4

1.8	Consider the following Verilog code snippet with initial values of "a=0" and "b=1" at "t=0":
	always
	begin
	#5; a =~a;
	b =~b;
	end
	What will be the values of "a" and "b" at "t=15"
		02	1	3

1.9	A flip-flops has a 3 ns delay. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency at which the counter can operate reliably?	02	2	3
1.10	If initial state of this right shift register is 1011, and serial data 10110011 is input to this. What is the output generated after 7 clock pulses?	02	3	3
1.11	A processor has 32 registers. How many bits are required to address a single register?	01	1	3
1.12	For the Adder/Subtractor circuit, if K=1, C_0=1, A=1010 and B=0011, find the output. 	02	2	3
1.13	In a 4 bit parallel adder ciruit to get increment A as output, the input to be given at B and Cin are	01	2	3

Part B

Question No	Question	M	CO	BT
2a	Write Truth table for the logic diagram shown in fig . Write Verilog code using data flow model. Assume a is vector input.	06	1	1

				
2b	Design Half adder circuit using primitive gates. Develop Full adder using half adders and write Verilog structural code for full adder using half adders.	10	2	3
3a	Design binary full adder circuit using 4:1 MUX. Obtain the logic expression and Draw the logic diagram.	08	4	3
3b	Implement the function F = Σ m(1,3,5,7) using a 3:8 decoder.	04	2	3
3c	Illustrate with an example the use of multiplexers in implementing Boolean logic functions.	04	1	3
	OR			
4a	Design a BCD to Excess-3 code converter using basic combinational blocks. Draw the truth table, derive simplified Boolean expressions by using K-maps and show how the logic can be mapped to decoders. Provide a Verilog behavioral model of the converter and discuss where such converters are used in real-time digital systems.	10	3	3
4b	Implement the following Boolean function using 8:1 MUX : F (A,B,C,D) = ∑m (0,1, 3, 4, 8,9,15)	06	2	3
5a	Design MOD 6 Synchronous Down counter using JK Flip Flops.	08	2	2
5b	For the state diagram shown in Fig, perform state reduction with the help for state table. Write reduced state diagram. If input to this machine is sequence 01010110100 then, what is the output sequence produced by this machine? 	08	4	3
	OR			
6a	A PN flip-flop has four operations: clear to 0, no change, complement, and set to 1, when inputs P and N are 00, 01, 10, and 11, respectively. i) Tabulate the characteristic table. ii) Derive the characteristic equation. iii) Tabulate the excitation table. iv) Show how the PN flip-flop can be converted to a D flip-flop.	08	3	3
6b	Derive the state table, state diagram and sequential circuit for the expression given below	08	2	3

$T_A = Bx$

$T_B = x$

$y = AB$

7a	Consider the case of a circuit to detect a pair of 1's or 0's in the single bit input. That is, input will be a series of one's and zero's. If two one's or two zero's comes one after another, output should go high. Otherwise output should be low. Develop a verilog Code for the circuit using Mealy FSM .	10	3	4
7b	Design 4 bit ripple counter using D flipflos. Write waveform and explain the counting opeartion	06	2	2
	OR			
8a	Draw the state diagram of a circuit that detects the following sequence: 10110. The detector must assert output 'z=1' when the sequence is detected.(With Mealy Overlap and non overlap Model).	08	2	3
8b	Differentiate between SISO, SIPO, PISO, and PIPO shift registers with examples	08	2	4
9a	Design and explain Processor unit employing a scratch pad memory with an example of micro-operation, explain in the sequence of execution.	06	3	3
9b	Design an Processor unit with two port scratchpad memorty. Illustrate with an example how microoperation is executed in one cycle.	10	4	4
	OR			
10	a) Design an arithmetic and logic unit using Full adder and gates to perform the operations shown shown in table. Obtain the expression for inputs Xi, Yi and Zi of full adder. Write the logic diagram for one stage.	16	4	4

Arithmetic Operations					Logical Operations			
S_2	S_1	S_0	$C_{in}=0$	$C_{in}=1$	S_2	S_1	S_0	$C_{in}=X$
0	0	0	F=A	F=A+1	1	0	0	OR
0	0	1	F=A+B	F=A+B+1	1	0	1	XOR
0	1	0	F=A-B-1	F=A-B	1	1	0	AND
0	1	1	F=A-1	F=A	1	1	1	Complement

b) Show how AND and OR operations are executed. Also show how addition and data transfer instructions are executed.